

R Basics Demographics & ANOVA

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Load Packages

```
library(tidyverse)
```

```
## — Attaching core tidyverse packages — tidyverse 2.0.0 —
## ✓ dplyr      1.1.4      ✓ readr      2.1.5
## ✓ forcats    1.0.0      ✓ stringr    1.5.1
## ✓ ggplot2    3.5.1      ✓ tibble     3.2.1
## ✓ lubridate  1.9.3      ✓ tidyr      1.3.1
## ✓ purrr      1.0.2
## — Conflicts — tidyverse_conflicts() —
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(psych)
```

```
##
## Attaching package: 'psych'
##
## The following objects are masked from 'package:ggplot2':
##
##      %+%, alpha
```

Let's first look at our demographics data.

Read in Demographics Data

```
demographics <- read.csv("demographics.csv")
head(demographics)
```

```
##      ID Initials Eighteen_or_Older Fluent_in_English CogPsych_Student Age Gender
## 1 P01          CW                Yes                Yes                Yes  19 Female
## 2 P02          IT                Yes                Yes                Yes  20 Female
## 3 P03          EL                Yes                Yes                Yes  20 Female
## 4 P04          SH                Yes                Yes                Yes  19 Female
## 5 P05          MD                Yes                Yes                Yes  21 Female
## 6 P06          JH                Yes                Yes                Yes  19   Male

##      Race_Ethnicity Academic_Year Psych_Major
## 1 White/Caucasian   Sophomore        Yes
## 2 White/Caucasian   Sophomore        Yes
## 3 White/Caucasian    Junior          No
## 4 White/Caucasian   Sophomore        Yes
## 5 White/Caucasian    Senior          Yes
## 6 White/Caucasian   Sophomore        No
```

```
nrow(demographics)
```

```
## [1] 50
```

```
ncol(demographics)
```

```
## [1] 10
```

Fifty rows and ten columns in this data frame.

Demographics, One Column at a Time

We will `group_by()` the “Eighteen_or_Older” column and use `summarise()` to get the percent of each answer option. We will do this for all of the various demographics columns.

```
demographics %>%
  group_by(Eighteen_or_Older) %>%
  summarise(percent = 100 * n() / nrow(demographics))
```

```
## # A tibble: 1 × 2
##   Eighteen_or_Older percent
##   <chr>             <dbl>
## 1 Yes              100
```

All fifty participants are eighteen or older.

```
demographics %>%
  group_by(Fluent_in_English) %>%
  summarise(percent = 100 * n() / nrow(demographics))
```

```
## # A tibble: 1 × 2
##   Fluent_in_English percent
##   <chr>             <dbl>
## 1 Yes              100
```

All fifty participants are fluent in English.

```
demographics %>%
  group_by(CogPsych_Student) %>%
  summarise(percent = 100 * n() / nrow(demographics))
```

```
## # A tibble: 1 × 2
##   CogPsych_Student percent
##   <chr>             <dbl>
## 1 Yes              100
```

All fifty participants are students in a cognitive psychology course.

```
demographics %>%
  group_by(Age) %>%
  summarise(percent = 100 * n() / nrow(demographics))
```

```
## # A tibble: 4 × 2
##   Age percent
##   <int>    <dbl>
## 1    19     34
## 2    20     40
## 3    21     22
## 4    22      4
```

34% of participants are 19 years old, 40% of participants are 20 years old, 22% of participants are 21 years old, and 4% of participants are 22 years old.

```
demographics %>%
  group_by(Gender) %>%
  summarise(percent = 100 * n() / nrow(demographics))
```

```
## # A tibble: 4 × 2
##   Gender      percent
##   <chr>        <dbl>
## 1 Female      62
## 2 Male        26
## 3 Non-binary   8
## 4 Transgender  4
```

62% of participants identify as female, 26% of participants identify as male, 8% of participants identify as non-binary, and 4% of participants identify as transgender.

```
demographics %>%
  group_by(Race_Ethnicity) %>%
  summarise(percent = 100 * n() / nrow(demographics))
```

```
## # A tibble: 5 × 2
##   Race_Ethnicity      percent
##   <chr>            <dbl>
## 1 Asian              24
## 2 Black/African American 10
## 3 Hispanic/Latino       2
## 4 Pacific Islander      6
## 5 White/Caucasian      58
```

24% of participants identify as Asian, 10% of participants identify as black or African American, 2% of participants identify as Hispanic or Latino, 6% of participants identify as Pacific Islander, and 58% of participants identify as white or Caucasian.

```
demographics %>%
  group_by(Academic_Year) %>%
  summarise(percent = 100 * n() / nrow(demographics))
```

```
## # A tibble: 3 × 2
##   Academic_Year percent
##   <chr>          <dbl>
## 1 Junior         38
## 2 Senior         20
## 3 Sophomore     42
```

38% of participants are juniors, 20% of participants are seniors, and 42% of participants are sophomores.

```
demographics %>%
  group_by(Psych_Major) %>%
  summarise(percent = 100 * n() / nrow(demographics))
```

```
## # A tibble: 2 × 2
##   Psych_Major percent
##   <chr>          <dbl>
## 1 No            22
## 2 Yes          78
```

22% of participants are not psychology majors and 78% of participants are psychology majors.

Now that we have looked at all the demographics data, let's look at final test performance and run our ANOVA. First, let's read in our data.

Read in Final Test Data

```
final_test <- read.csv("final_test.csv")
head(final_test)
```

```
##      ID Initials Group Test1 Test2 Test3 Test4 Test5 Test6 Test7 Test8 Test9
## 1 P01      CW   One    1     1     1     1     2     2     3     4     2
## 2 P02      IT   Two    3     1     1     1     1     1     1     1     1
## 3 P03      EL   One    2     1     4     3     4     3     4     2     3
## 4 P04      SH   Two    2     2     1     2     1     3     4     4     4
## 5 P05      MD   One    1     2     1     1     1     1     3     3     3
## 6 P06      JH   Two    1     3     1     4     1     2     1     1     1
##      Test10
## 1         2
## 2         1
## 3         4
## 4         4
## 5         4
## 6         1
```

```
nrow(final_test)
```

```
## [1] 50
```

```
ncol(final_test)
```

```
## [1] 13
```

Fifty rows and thirteen columns in this data frame. All our participants match up by their “ID” and “Initials.” Participants are either in “Group One” or “Group Two” and answered ten total test questions.

Pivot Data to Long Format for Analysis

```
final_test %>% pivot_longer(cols=c(Test1:Test10),
                           names_to="Question",
                           values_to="Response") -> final_test_long
```

Let's check that everything looks right.

```
head(final_test_long)
```

```
## # A tibble: 6 × 5
##   ID      Initials Group Question Response
##   <chr> <chr>      <chr> <chr>      <int>
## 1 P01    CW        One   Test1        1
## 2 P01    CW        One   Test2        1
## 3 P01    CW        One   Test3        1
## 4 P01    CW        One   Test4        1
## 5 P01    CW        One   Test5        2
## 6 P01    CW        One   Test6        2
```

```
nrow(final_test_long)
```

```
## [1] 500
```

Now we have 500 rows, 10 for each participant (50 x 10 = 500).

We can now also see the responses for each of the ten test questions. 1 indicates the correct answer (aka answer option “a”). 2, 3, and 4 are wrong answers (aka answer options “b”, “c”, and “d”). Let’s code that (what is right and what is wrong).

```
final_test_long %>% mutate(Correct = ifelse(Response == 1, 1, 0)) -> final_test_long
```

Let’s check that everything looks right.

```
head(final_test_long)
```

```
## # A tibble: 6 × 6
##   ID      Initials Group Question Response Correct
##   <chr> <chr>      <chr> <chr>      <int>    <dbl>
## 1 P01    CW        One   Test1        1        1
## 2 P01    CW        One   Test2        1        1
## 3 P01    CW        One   Test3        1        1
## 4 P01    CW        One   Test4        1        1
## 5 P01    CW        One   Test5        2        0
## 6 P01    CW        One   Test6        2        0
```

Yep, everything looks great. Let’s first take a look at our descriptive statistics (i.e., means and standard deviations) before we run our ANOVA.

```
final_test_long %>% group_by(Group) %>% summarise(accuracy = mean(Correct),
                                                    SD = sd(Correct))
```

```
## # A tibble: 2 × 3
##   Group accuracy    SD
##   <chr>    <dbl> <dbl>
## 1 One      0.604 0.490
## 2 Two      0.696 0.461
```

Group One has a mean accuracy of 0.60 (~60%) and a standard deviation of 0.49. Group 2 has a mean accuracy of 0.70 (~70%) and a standard deviation of 0.46.

Now let's run our ANOVA.

```
model <- aov(Correct ~ Group, data = final_test_long)
summary(model)
```

```
##           Df Sum Sq Mean Sq F value Pr(>F)
## Group      1   1.06   1.0580   4.675 0.0311 *
## Residuals 498 112.69   0.2263
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

We can see that there is a significant difference between our Groups. Group 2 performed significantly better on the final test compared to Group 1 ($p = 0.03$ *).